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Optically pumped polarized H^- ion source for RHIC spin physics

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A new optically pumped polarized H^- ion source (OPPIS) was developed for the RHIC polarization program and successfully used for the first polarized beam commissioning at RHIC. The OPPIS produces in excess of 1.0 mA H^- ion current at about 80% polarization. An ECR primary proton source development and a new 29 GHz microwave power supply are described. A new type sodium-jet ionizer cell is biased to -32 kV to produce a 35 keV polarized beam ready for injection to the RFQ. Higher current and higher polarization were also obtained with the biased jet-cell in comparison with an old oven-type ionizer cell. © 2002 American Institute of Physics. [DOI: 10.1063/1.1427669]

I. INTRODUCTION

Collider experiments at RHIC at up to $\sqrt{s}=500$ GeV energy and $2 \times 10^{32} \text{ cm}^{-2} \text{ s}^{-1}$ luminosity will provide unique complimentary information on proton structure, QCD, and electroweak interaction of quarks.^{1,2} To meet the required luminosity the intensity must be 2×10^{11} protons per bunch in RHIC. The intensity of the polarized H^- ion source has to be about 9×10^{11} H^- ions/pulse, which corresponds to a 150 mA μs integral intensity (e.g., a 300 μs pulsed current of 0.5

mA). Such an intensity can be easily obtained in a dc operation and up to 10.0 mA in a pulsed operation.^{3,4} The optically pumped polarized H^- ion source (OPPIS) technique is based on spin-transfer collisions between a primary proton or atomic hydrogen beam of a few keV energy and optically pumped alkali-metal vapors. The RHIC OPPIS was developed in a collaboration among Brookhaven National Laboratory (BNL), KEK, INR (Moscow), and TRIUMF. It is based on the KEK OPPIS which was successfully used for production of a vector polarized D beam at KEK and upgraded at TRIUMF for high current H^- operation.⁵ The new

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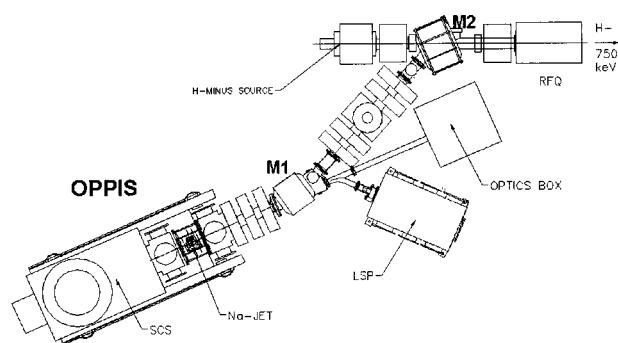


FIG. 1. A general polarized RHIC injector layout. ECR: electron-cyclotron resonance proton source; SCS: superconducting solenoid; Na-jet: sodium-jet ionizer cell; LSP: Lamb-shift polarimeter; M1, M2: dipole bending magnets.

OPPIS routinely produces 0.5 mA current in a 300 μ s pulse, which can be expanded to 500 μ s—about the maximum linac pulse duration. This intensity is 30 times higher than the typical intensity of the old BNL atomic beam source and sufficient to produce the required RHIC bunch intensity with a single source pulse.

Polarized H^- ions are produced in the OPPIS at 35 keV beam energy. The beam is accelerated to 200 MeV with a RFQ and linac for strip injection to the booster. About 50% of the OPPIS beam intensity can be accelerated to 200 MeV. The 300 μ s H^- ion pulse is captured in a single booster bunch which contains about 4×10^{11} polarized protons. The single bunch is accelerated in the booster to 1.5 GeV kinetic energy and then transferred to the AGS, where it is accelerated to 25 GeV for injection to RHIC.

The OPPIS initial longitudinal polarization is converted to the transverse direction while the beam passes two bending magnets (see Fig. 1). The second 47.4° bending magnet switches linac injection between polarized and unpolarized high intensity (up to 100 mA) H^- ion beam. The magnet is pulsed and either beam can be accelerated pulse to pulse in the same RFQ. A pulsed focusing solenoid in front of the RFQ is tuned for the optimal for either beam transmission. It rotates the polarization direction for about 420°, but still keeps it in the transverse plane, and a final polarization alignment to the vertical direction can be adjusted by a spinrotator solenoid in the 750 keV beam transport line before injection to the linac. The AGS cycle for polarized beam operation is 3 s. The OPPIS operates at 1 Hz repetition rate and two additional source pulses can be directed to the 200 MeV p -carbon polarimeter for polarization monitoring by another pulsed bending magnet in the high-energy beam transport line. The polarization alignment to the vertical plane is done using a 200 MeV polarimeter.

II. ECR PRIMARY PROTON SOURCE UPGRADE

The OPPIS current is determined by the ECR primary proton source, since the optimal neutralization efficiency in the optically pumped Rb vapor is 50%–70%, the sodium ionizer efficiency is constant (about 9% at 3.0–4.0 keV beam energy), and the ionizer diameter is limited by emittance

specification. The OPPIS ECR source operation is quite different from conventional ECR ion sources developed for multiply charged heavy ion production.

Obtaining a high polarization requires a magnetic field (in excess of 25 kG) in the optically pumped cell and ion extraction region. In the KEK OPPIS the protons were produced in a 6.4 kG resonance field and extracted at a 27 kG field. With the 28 GHz microwave power supply at TRIUMF the resonance frequency is 10 kG. This gives a factor of 2–3 current gain, other conditions being similar. In the BNL OPPIS the ECR ion source operates at a 29 GHz frequency. It is situated in the superconducting solenoid having three separately adjusted coils, which produce a 10 kG resonant field for ECR source operation and up to 27 kG flat top field for the optically pumped Rb vapor cell. The 29 GHz microwave power of about 0.8 kW, required for saturating the proton current yield, is produced by a cw extended interaction oscillator (CPI VKQ2453L1 maximum power up to 1.2 kW). Two of these old 28 GHz tubes were originally acquired back in 1983 and were used at TRIUMF for more than 15 years and more than 10 000 h operation time each. For the BNL OPPIS a 28–35 GHz gyrotron tube was considered as an option. The cost of gyrotron (10 kW cw) and EIO tubes are similar but the gyrotron power supply would be more powerful and expensive. An important factor was also to keep the microwave generators at TRIUMF and BNL similar, to allow tube exchange if necessary. A power supply for the tube was built at BNL on the base of 20 kV, 500 mA switching PS (Universal Voltronics BRS-10000). Similar to the TRIUMF OPPIS prototype, the BNL OPPIS is practically a dc source, except that it has 35 keV pulsed beam acceleration, and optical pumping is by a pulsed laser. Therefore, the current pulses are very stable, flat, and pulse duration can be extended if necessary. The microwave power can be pulsed by anode voltage modulation (a Behlke HTS-300 switch is used for this purpose). The pulsed ECR source operation was tested at 1 Hz repetition rate and a pulse duration 10–100 ms. It was observed that about 10%–20% higher current can be obtained in a pulsed operation at the front of the pulse with about 50% less hydrogen supply to the discharge tube. Perhaps hydrogen desorption from the quartz liner produces sufficient gas density at the pulse front, which allows one to reduce the hydrogen load to the vacuum pumps and improve the residual vacuum in the Rb cell. This also reduces the depolarization due to neutralization in the residual gas.

The OPPIS ion extraction system consists of three 1.0-mm-thick planar molybdenum electrodes spaced 1.0 mm apart, having 0.95 diameter holes in a hexagonally close-packed configuration with 1.15 mm distance between centers. Precise manufacturing (by EDM technique) and alignment of multihole electrodes is required for the highest performance and lifetime. In dc operation the grids are heated by the beam and cooling is necessary to prevent melting of the Delrin insulator. The grids are located about 3.0 cm from the high density Rb vapor cell, and to prevent Rb condensation the grids are mounted to stainless steel, thin wall holding plates, which allows a large temperature gradient from the hot grids to the outside diameter, where the cooling lines are attached. In pulsed operation the cooling is

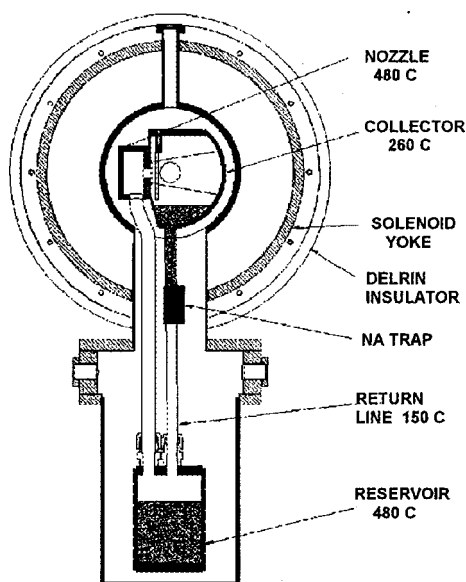


FIG. 2. The sodium-jet ionizer cell. The cell is situated inside a solenoid of 1.4 kG field strength.

turned off and the ECR is switched to dc operation twice a day to clean the grids.

The proton beam produces positive Rb^+ ions when it captures electrons from Rb atoms. These ions are confined by the high solenoid field and can travel only longitudinally out of the cell. The Rb ion flow to the extraction grid is suppressed by a positive voltage of about 100 V, which is applied to the third grid, and Rb ions travel downstream to negatively biased deflecting plates. In the high current source, the primary proton current is about 80 mA, and when 50% of the beam is neutralized a 40 mA Rb ion beam is produced in dc operation. This ionization causes more Rb losses than thermal vapor diffusion out of the source. In pulsed operation the average current is greatly reduced, and Rb consumption is reduced.

III. SODIUM-JET IONIZER CELL OPERATION

The neutral atomic hydrogen beam of 3.0 keV energy enters the ionizer cell, and H^- ions produced in the cell can then be accelerated to 35.0 keV (which is required for injection to RFQ) by ionizer biasing to -32.0 kV. A large Na cell aperture is essential for this purpose because the neutral beam, collimated to 2.0 cm in diameter before the cell, should not touch the biased cell parts, otherwise the secondary emission will cause sparking. A new jet-type ionizer cell with transverse sodium flow was developed to allow large apertures (see Fig. 2).⁴ The reservoir is loaded with 150 g of sodium metal through the loading port at the top of the cell. A loading cylinder is filled with a solid piece of sodium in the glove box. A transport stainless steel tube of a 0.5 in. diameter goes through the solenoid yoke down to the bottom of the collector and is sealed by swagelock coupling to the ionizer cell vacuum chamber. When the cylinder is heated up to about 150°C liquid sodium flows down to the collector and further to the reservoir through the return line (the reservoir is heated to about 300°C). This procedure can be repeated, if necessary, for long-term operation.

The reservoir operational temperature is about 480°C . At this temperature the sodium vapor density is about 10^{17} atoms/ cm^3 . The vapor is delivered through a hot transport line to the nozzle, which produces a horizontal vapor jet having an effective thickness of about 5×10^{14} atoms/ cm^2 , sufficient for H^- yield saturation. The nozzle slit is 3.0 mm wide and 20 mm tall. The transport tube and nozzle are heated to 480°C . The sodium vapor condenses on the collector walls, which are cooled to 120°C (above melting point) by hot water circulation. The hot sodium vapor sticking efficiency to the collector wall depends on the wall surface condition. A large sodium flow increase was observed from the collector walls coated by sodium oxide layers after venting of the operational cell to atmosphere. The ventilation should be done with great care using a dry nitrogen or argon in a cold cell, otherwise cell cleaning is required. Liquid sodium flows down the return tube and back to reservoir. The return tube temperature is kept at 150°C by an attached cooling line with controlled hot water flow. Backstreaming vapor flow through the return line is negligible due to the low conductance at 150°C . Sodium in the jet cell circulates along the path reservoir–nozzle–collector–return line–reservoir and the system provides continuous, stable operation for hundreds of hours with 150 g sodium. Without the circulation the cell works for only 3 h, measured in a test with the collector cooled to 25°C . The frozen sodium in the collector had a volcano shape perfectly confined within the 10-cm-long collector. While a large amount of sodium is circulated inside the cell, the sodium flow outside is much less than with the oven-type cell.

The whole ionizer assembly, including the solenoid magnet, is electrically isolated from the rest of the source by 50-mm-thick Delrin flanges. The acceleration to 35 keV is produced in a two gap extraction system of 25 and 75 mm lengths (see Fig. 2). The extraction voltage applied across the first gap is 4.0 kV and across the second gap is 28.0 kV. The immediate acceleration reduces polarized current losses and improves polarization, since the energy for the spin-transfer collisions can be set optimal for efficient polarization transfer and ionization.

IV. EXPERIMENTAL RESULTS

The final tests at TRIUMF were performed with the TRIUMF Oxford made superconducting solenoid. The proton polarization was optimized by using a Lamb-shift polarimeter and measured after acceleration to 300 keV with a nuclear scattering polarimeter based on the $\text{Li}^6(p, \text{He}^3)\text{He}^4$ reaction. At low Rb thickness a polarization of 85% was achieved in dc mode with a cw Ti:sapphire laser. At higher Rb density polarization drops, but the use of a pulsed Cr:LiSAF laser restores the polarization to 85% at 1.6 mA H^- ion beam intensity (see Fig. 3). A long pulse, flash lamp pumped Cr:LiSAF laser had been developed for BNL OP-PIS. This laser can produce in excess of 1.0 kW power with 400 μs pulse duration.

The OPPIS was shipped to BNL in September of 1999, installed at the 200 MeV linac injector, and used for the RHIC spin commissioning in August–September of 2000.

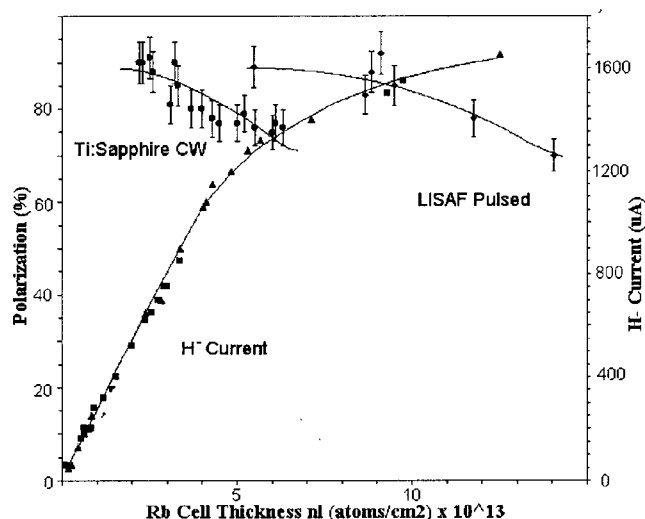


FIG. 3. The H^- beam current and polarization vs the optically pumped Rb vapor thickness.

The OPPIS worked continuously and reliably during the three week final run. So far, a maximum current of 1.1 mA has been obtained, as measured at 35 keV in a Faraday cup 2 m from the source exit. About 50% of this current was accelerated to 200 MeV (see Fig. 4). The source current is about 70% of the current obtained at the tests with the TRIUMF solenoid. The Oxford solenoid has a cold yoke, in contrast with the KEK solenoid which has a large hole in the room temperature yoke. This disturbs the magnetic field symmetry and introduces a transverse field, which missteers the low energy proton beam. The displacement was observed by direct measurements of the atomic H beam profile at the entrance of the ionizer cell. The Lamb-shift polarimeter has been used for initial polarization measurements and polarization optimization. For these measurements the H^- beam was converted to a proton beam in a pulsed helium ionizer cell. In another mode of operation, the polarization of the protons produced together with the H^- ions in the sodium ionizer

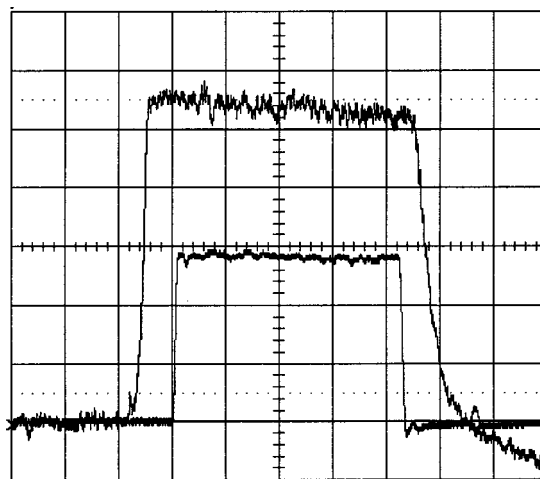


FIG. 4. The polarized H^- ion beam current pulse: top trace—the OPPIS beam current at 35 keV energy; bottom trace—the 200 MeV beam current. A vertical scale is 0.2 mA/div, a horizontal scale is 100 μ s/div.

cell was measured. Finally, the polarization was measured in the inclusive p -carbon polarimeter at 200 MeV, which was recently calibrated by comparison with elastic proton–deuteron scattering. About 75% polarization has been achieved so far at BNL. Work is in progress to identify possible polarization losses in the source, LEPT, and spin rotating magnets.

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